

National Coverage Determination (NCD)

Transcutaneous Electrical Nerve Stimulation (TENS) for Chronic Low Back Pain (CLBP)

160.27

Links in PDF documents are not guaranteed to work. To follow a web link, please use the MCD Website.

Tracking Information

Publication Number

100-3

Manual Section Number

160.27

Manual Section Title

Transcutaneous Electrical Nerve Stimulation (TENS) for Chronic Low Back Pain (CLBP)

Version Number

1

Effective Date of this Version

06/08/2012

Implementation Date

01/07/2013

Description Information

Benefit Category

Durable Medical Equipment

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

The TENS is a type of electrical nerve stimulator that is employed to treat chronic intractable pain. This stimulator is attached to the surface of the patient's skin over the peripheral nerve

to be stimulated. It may be applied in a variety of settings (in the patient's home, a physician's office, or in an outpatient clinic). Payment for TENS may be made under the durable medical equipment benefit.

A. General

For the purposes of this decision chronic low back pain (CLBP) is defined as:

1. an episode of low back pain that has persisted for three months or longer; and
2. is not a manifestation of a clearly defined and generally recognizable primary disease entity. For example, there are cancers that, through metastatic spread to the spine or pelvis, may elicit pain in the lower back as a symptom; and certain systemic diseases such as rheumatoid arthritis and multiple sclerosis manifest many debilitating symptoms of which low back pain is not the primary focus.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective June 8, 2012, the Centers for Medicare & Medicaid Services (CMS) will allow coverage for Transcutaneous Electrical Nerve Stimulation (TENS) for CLBP only when all of the following conditions are met.

In order to support additional research on the use of TENS for CLBP, we will cover this item under section 1862(a)(1)(E) of the Social Security Act (the Act) subject to all of the following conditions:

1. Coverage under this section expires three years after the publication of this decision on the CMS website.
2. The beneficiary is enrolled in an approved clinical study meeting all of the requirements below. The study must address one or more aspects of the following questions in a randomized, controlled design using validated and reliable instruments. This can include randomized crossover designs when the impact of prior TENS use is appropriately accounted for in the study protocol.
 - i. Does the use of TENS provide clinically meaningful reduction in pain in Medicare beneficiaries with CLBP?
 - ii. Does the use of TENS provide a clinically meaningful improvement of function in Medicare beneficiaries with CLBP?
 - iii. Does the use of TENS impact the utilization of other medical treatments or services used in the medical management of CLBP?

These studies must be designed so that the patients in the control and comparison groups receive the same concurrent treatments and either sham (placebo) TENS or active TENS intervention.

The study must adhere to the following standards of scientific integrity and relevance to the Medicare population:

- a. The principal purpose of the research study is to test whether a particular intervention potentially improves the participants' health outcomes.
- b. The research study is well supported by available scientific and medical information or it is intended to clarify or establish the health outcomes of interventions already in common clinical use.
- c. The research study does not unjustifiably duplicate existing studies.
- d. The research study design is appropriate to answer the research question being asked in the study.
- e. The research study is sponsored by an organization or individual capable of executing the proposed study successfully.
- f. The research study is in compliance with all applicable Federal regulations concerning the protection of human subjects found at 45 CFR Part 46. If a study is regulated by the Food and Drug Administration (FDA), it must be in compliance with 21 CFR parts 50 and 56.
- g. All aspects of the research study are conducted according to appropriate standards of scientific integrity (see <http://www.icmje.org>).
- h. The research study has a written protocol that clearly addresses, or incorporates by reference, the standards listed here as Medicare requirements for CED coverage.
- i. The clinical research study is not designed to exclusively test toxicity or disease pathophysiology in healthy individuals. Trials of all medical technologies measuring therapeutic outcomes as one of the objectives meet this standard only if the disease or condition being studied is life threatening as defined in 21 CFR § 312.81(a) and the patient has no other viable treatment options.
- j. The clinical research study is registered on the ClinicalTrials.gov website by the principal sponsor/investigator prior to the enrollment of the first study subject.
- k. The research study protocol specifies the method and timing of public release of all prespecified outcomes to be measured including release of outcomes if outcomes are negative or study is terminated early. The results must be made public within 24 months of the end of data collection. If a report is planned to be published in a peer reviewed journal, then that initial release may be an abstract that meets the requirements of the International Committee of Medical Journal Editors (<http://www.icmje.org>).
- l. The research study protocol must explicitly discuss subpopulations affected by the treatment under investigation, particularly traditionally underrepresented groups in clinical studies, how the inclusion and exclusion criteria effect enrollment of these populations, and a plan for the retention and reporting of said populations on the trial. If the inclusion and exclusion criteria are expected to have a negative effect on the recruitment or retention of underrepresented populations, the protocol must discuss why these criteria are necessary

- m. The research study protocol explicitly discusses how the results are or are not expected to be generalizable to the Medicare population to infer whether Medicare patients may benefit from the intervention. Separate discussions in the protocol may be necessary for populations eligible for Medicare due to age, disability or Medicaid eligibility.

C. Nationally Non-Covered Indications

TENS is not reasonable and necessary for the treatment of CLBP under section 1862(a)(1)(A) of the Act.

D. Other

See [§160.13](#) for an explanation of coverage of medically necessary supplies for the effective use of TENS. See [§160.7.1](#) for an explanation of coverage for assessing patients suitability for electrical nerve stimulation therapy. See [§10.2](#) for an explanation of coverage of transcutaneous electrical nerve stimulation (TENS) for acute post-operative pain. Please note, [§280.13](#) Transcutaneous Electrical Nerve Stimulators (TENS) NCD has been removed from the NCD manual and incorporated into NCD [160.27](#)

(This NCD last reviewed June 2012.)

Cross Reference

[CED page](#)

Claims Processing Instructions

[TN 2511 \(Medicare Claims Processing\)](#)

[TN 2605 \(Medicare Claims Processing\)](#)

Transmittal Information

Transmittal Number

149

Coverage Transmittal Link

<http://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R149NCD.pdf>

Revision History

10/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the

result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11083](#)) (CR12482)

01/2018 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 2005](#)) (CR10318)

11/2017 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 1975](#)) (CR10318)

05/2014 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 10/06/2014 Effective date: 10/1/2015. ([TN 1388](#)) ([TN 1388](#)) (CR 8691)

The purpose of this Change Request (CR) is to inform contractors that effective for claims with dates of service on or after June 8, 2012, Medicare will only allow coverage of TENS for CLBP defined for this decision as pain for more than 3 months and not a manifestation of a clearly defined and generally recognizable primary disease entity, when the patient is enrolled in an approved clinical study under coverage with evidence development (CED). ([TN 144](#)) (CR7836)

NOTE: This instruction is being re-issued due to original transmission erroneously omitted Section 280.13 from transmittal page and Part A. Section 160.7.1 of the manual.

Transmittal 144, dated August 3, 2012, is being rescinded and replaced by Transmittal 149 dated November 30, 2012. The NCD Manual is being revised to include the new NCD section 160.27 and revised sections as a cross reference in all existing TENS manual

sections. The Transmittal Number, date issued and all other information remains the same.
([TN 149](#)) (CR7836)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Transcutaneous Electrical Nerve Stimulation for Chronic Low Back Pain \(CAG-00429N\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Transcutaneous Electrical Nerve Stimulation (TENS) for Chronic Low Back Pain (CLBP)	1	06/08/2012 - N/A	You are here